DISTRIBUTION-SHIFTED SYNTHETIC MULTITRACK PRETRAINING FOR LARGE-VOCABULARY AUDIO CHORD RECOGNITION

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ABSTRACT

Audio chord recognition (ACR) models trained on human-annotated corpora face a persistent triadic performance plateau: canonical benchmarks concentrate over 80% of duration in major and minor triads, leaving rare triads (`sus2`, `sus4`, `dim`, `aug`) and upper extensions (7th–13th) acoustically sample-starved. Consequently, neural classifiers collapse into majority-class predictors, and inverse-frequency loss reweighting fails to compensate for the physical absence of diverse training exemplars. We propose a Sim-to-Real pretraining framework driven by a probabilistic composition generator that synthesizes $5{,}000$ multitrack audio tracks (460.3 h) using rare-chord quota scheduling, physically constrained multi-instrument voicings, and stochastic random-mix acoustic augmentation. We pretrain ChordFormer—a 4-block Conformer encoder with six decomposed output heads—on synthetic multitracks and finetune on real audio via inverted discriminative learning rates ($\mathbf{s}_{\text{lr}} = [0.2, 0.5, 1.0]$) and floor curriculum replay ($\rho_{\min} = 0.10$). Across all $11{,}436{,}357$ frames of the $1{,}217$-song benchmark, our curriculum achieves best-in-class recall on underrepresented harmonies: boosting rare triad recall from 41.67% to 48.31% ($+6.64$ pp; F_1 from 53.11% to 60.94%), 9th extensions from 35.33% to 43.58% ($+8.25$ pp), natural 9ths by $+25.16$ pp ($31.64\% \rightarrow 56.80\%$), altered `#9` by $79\times$ ($0.07\% \rightarrow 5.53\%$), and flat 13ths by $7.8\times$ ($0.84\% \rightarrow 6.58\%$). While duration-weighted MIREX WCSR shifts from 81.49% to 79.59% due to metric bias toward dominant triads, cross-fold variance contracts across all ten evaluation rules.

1. INTRODUCTION

Automatic Chord Recognition (ACR) underpins computational musicology, structural segmentation, and content-based audio indexing [1]. Although ACR architectures have progressed from handcrafted chroma templates and hidden Markov models to convolutional-recurrent networks and self-attention Transformers [2–4], performance remains bounded by an empirical triadic plateau: classifiers reach 82%–84% accuracy on major and minor triads but degrade sharply on four-note tetrads and extended harmonies [1, 5].

This ceiling stems not from insufficient model capacity, but from the severe Zipfian skew of real-world musical corpora [5, 6]. In the canonical $1{,}217$-song benchmark compiled by Humphrey and Bello [5] (combining McGill Billboard [6], Isophonics [7], and USPop2002), major (65.75%) and minor (22.16%) triads occupy 87.91% of non-silent duration (73.7 h, $11{,}436{,}357$ frames). Rare triads (`sus4`, `dim`, `sus2`, `aug`) collectively span under 3.5% of duration, while altered upper extensions (`#11` at 0.13%, `b9` at 0.13%, `b13` at 0.08%) inhabit a sparse tail. Moreover, human inter-annotator disagreement exceeds 40% on inversions and upper extensions [8], compounding sample scarcity with label noise.

Recent architectures address large-vocabulary recognition via structured multi-task losses [9], 6-head chord decomposition [10, 11], and contrastive pseudo-labeling [12]. Specifically, Jiang et al. [10] and Akram et al. [11] factorize chords into six categorical heads (root+triad, bass, and 7th, 9th, 11th, 13th degrees) trained under inverse-frequency reweighted cross-entropy. Yet loss reweighting cannot overcome an absence of physical acoustic exemplars: when an altered degree occurs in only a few dozen frames across a corpus, scaling its gradient penalty magnifies gradient variance and overfits to the idiosyncratic instrumentation and room acoustics of those few tracks. Resolving the long tail requires intervening directly on the training audio distribution.

We propose a Sim-to-Real pretraining framework driven by procedurally synthesized multitrack audio, contributing: (1) **Probabilistic Composition Generator**, an open-source engine sampling progressions along transitions from ChoCo [13], McGill Billboard [6], and Weimar Jazz [14] under deterministic quota scheduling (upweighting rare triads up to $15.6\times$ and altered extensions up to $12.3\times$), enforcing physical register bounds and 100% extension sounding across four stems with random-mix augmentation to render $5{,}000$ tracks (460.3 h) and frame-exact Harte labels [15]; (2) **Two-Stage Sim-to-Real Curriculum**, a transfer regimen for ChordFormer [11] combining synthetic pretraining with real-audio finetuning via inverted discriminative learning rates ($\mathbf{s}_{\text{lr}} = [0.2, 0.5, 1.0]$) and floor curriculum replay ($\rho(\eta) = 0.150 \rightarrow 0.100$); and (3) **Multi-Head and WCSR Benchmark Analysis**, a 5-fold cross-validated evaluation over all $11{,}436{,}357$ frames demonstrating best-in-class rare-chord recall ($+6.64$ pp on

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rare triads, $+25.16$ pp on natural 9ths, $79\times$ on `#9`, $7.8\times$ on `b13`) alongside an analysis of precision-recall trade-offs, duration-weighted WCSR bias, and polyphonic overtone masking.

2. RELATED WORK

2.1 Large-Vocabulary and Structured Chord Recognition

Early ACR systems restricted classification to 24 major/minor triads using chroma vectors and hidden Markov models [1, 2]. To scale to larger vocabularies without combinatorial explosion, McFee and Bello [9] formulated a structured multi-task loss over 170 classes (14 qualities), regularizing predictions via root, bass, and pitch-class targets; however, their taxonomy maps all upper extensions (9th, 11th, 13th) onto basic triads or tetrads.

Jiang et al. [10] preserved upper tertian degrees by decomposing chords into six categorical heads (Root+Triad, Bass, 7th, 9th, 11th, 13th) trained with CRNNs, inverse-frequency class reweighting, and CRF decoding. Self-attention architectures further improved temporal context modeling: the Bidirectional Transformer for Chords (BTC) [3] replaced recurrent layers with bidirectional self-attention, and Kim and Park [16] introduced event-level boundary pooling to suppress frame oversegmentation on non-triad harmonies.

Akram et al. [11] proposed ChordFormer, pairing Macaron-style Conformer blocks [4] with the 6-head decomposition of Jiang et al. [10] to capture local harmonic partials via depthwise convolution and global phrase structure via multi-head attention. Li et al. [12] applied contrastive learning and Noisy Student pseudo-labeling to unannotated audio across 170 classes. However, teacher-student pseudo-labeling suffers from confirmation bias on the harmonic tail: because teacher networks exhibit near-zero recall on rare alterations, pseudo-labels reinforce majority diatonic priors. Our approach instead supplies noise-free, symbolically verified ground-truth audio across the full extension vocabulary.

2.2 Synthetic Audio Datasets in MIR

Synthetic multitrack rendering offers a scalable remedy for annotation scarcity in MIR. In source separation, Slakh2100 [17] rendered 2,100 MIDI arrangements through virtual instrument libraries to provide isolated stems. For chord recognition, Ostermann et al. [18] developed Artificial Audio Multitracks (AAM), comprising 3,000 FluidSynth-rendered tracks generated via algorithmic composition in JFugue.

Majchrzak and Mańdziuk [19] benchmarked synthetic pretraining on BTC and Harmony Transformer models using AAM, showing consistent accuracy gains on major/minor pop progressions after real-audio finetuning, but zero transfer to extended chord vocabularies and degraded accuracy on jazz repertoires. Because AAM samples progressions exclusively from unextended diatonic triads (I–vi) [18], it replicates the triadic skew of real corpora. In contrast, our generator acts as an explicit distributional intervention: upweighting rare triads and altered extensions via quota scheduling, enforcing physical voicings so labeled extensions sound acoustically, and applying random-mix augmentation to bridge the SoundFont-to-studio gap.

3. PROBABILISTIC COMPOSITION GENERATOR

3.1 Distribution-Shifted Harmonic Sampling

Whereas AAM [18] renders 3,000 tracks from random diatonic triads (I–vi), omitting vii°, suspended/augmented triads, and 7th–13th extensions [19], we design `target_corpus_gen.py` as a distributional intervention over 5,000 tracks (3,250 pop/rock, $r_{\text{pop}} = 0.65$; 1,750 jazz, 0.35; 500,000 events, 100/track) along transitions extracted via ChoCo [13] from McGill Billboard [6] and OMRAS2 [7] (1,114 songs, 139,078 events) plus the Weimar Jazz Database [14] and iReal Pro (2,477 charts, 161,022 events).

We parameterize each event C_t in the 6-head space of Jiang et al. [10] and ChordFormer [11]: root $r_t \in \mathbb{Z}_{12}$ and triad $q_t \in \{\texttt{maj}, \texttt{min}, \texttt{sus4}, \texttt{sus2}, \texttt{dim}, \texttt{aug}, 5, 1\}$ form $S_t = (r_t, q_t)$ ($\hat{\mathbf{z}}_{0,t} \in \mathbb{R}^{97}$ with `N`); bass $b_t = (r_t + \Delta_{\text{bass},t}) \bmod 12$ for $\Delta_{\text{bass},t} \in \mathbb{Z}_{12}$ ($\hat{\mathbf{z}}_{1,t} \in \mathbb{R}^{13}$); and $\mathbf{E}_t = (e_{7,t}, e_{9,t}, e_{11,t}, e_{13,t})$ spans $\{\texttt{N}, 7, \texttt{b7}, \texttt{bb7}\}$ ($\hat{\mathbf{z}}_{2,t} \in \mathbb{R}^4$, mapping `dim7` and `maj6`/`min6` to `bb7`), $\{\texttt{N}, 9, \texttt{\#9}, \texttt{b9}\}$ ($\hat{\mathbf{z}}_{3,t} \in \mathbb{R}^4$), $\{\texttt{N}, 11, \texttt{\#11}\}$ ($\hat{\mathbf{z}}_{4,t} \in \mathbb{R}^3$), and $\{\texttt{N}, 13, \texttt{b13}\}$ ($\hat{\mathbf{z}}_{5,t} \in \mathbb{R}^3$), totaling 124 logits. Per song (4/4 meter), we sample tonic $k \sim U_{\mathbb{Z}}(0, 11)$ (initializing S_1 to `0_major`/`0_minor`), BPM $\sim U_{\mathbb{Z}}(60, 180)$, and beat durations $d_t \in \{8, 6, 4, 3, 2, 1.5, 1\}$ (`ww`–`q`) with pop/rock weights $(.02, .03, .60, .05, .20, .05, .05)$ and jazz weights $(.01, .01, .30, .08, .40, .05, .15)$ ($\Delta t_{\text{sec}} = 60 d_t/\text{BPM}$).

In `chord_gen.py`, sequence transitions factorize as:

$$P(C_t \,|\, C_{t-1}, C_{t-2}) = P(S_t \,|\, S_{t-1}, S_{t-2})\, P(b_t \,|\, r_t, b_{t-1}) \cdot P(\mathbf{E}_t \,|\, S_t, b_t, \mathbf{E}_{t-1}). \tag{1}$$

Stage 1 interpolates order-2, order-1, and unigram tables via Witten-Bell weights $\lambda_i = n_i/(n_i + D)$ ($D = 25.0$):

$$P_{\text{final}}(S_t) = \lambda_2 P_{\text{L2}}(S_t \,|\, S_{t-1}, S_{t-2}) + (1 - \lambda_2) \cdot \big[\lambda_1 P_{\text{L1}}(S_t \,|\, S_{t-1}) + (1 - \lambda_1) P_{\text{L0}}(S_t)\big], \tag{2}$$

using tempered $P_{\text{L1}} \propto P_{\text{raw}}^{1/\tau_1}$ ($\tau_{1,\text{pop}} = 2.2$, $\tau_{1,\text{jazz}} = 2.0$), floor $(1 - \epsilon)P_{\text{final}} + \epsilon P_{\text{L0}}$ ($\epsilon = 0.02$), and repeat discount $\delta = 0.20$ on $S_t = S_{t-1}$. Stage 2 applies order-1 bass backoff ($D = 25.0$), adds whole/half-note slash inversions at $p_{\text{inv}} = 0.10$ (0.60 3rd, 0.40 5th), and slices sub-measure passing bass motion. Stage 3 walks a 4-slot prefix trie across state Trie A ($N \geq 200$), triad Trie B ($N \geq 500$), and genre Trie C with $(b_t, \mathbf{E}_{t-1})$ backoff, merges enharmonic collisions (`bb7+13`, `sus4+11`, `sus2+9`, `dim+#11`, `aug+b13`), and injects $p_{\text{NC}} = 0.01$ silence (`N`) solely on unextended root-position `maj`/`min` triads.

We partition tracks ($r_{\text{nat}} = 0.50$) into 2,500 natural and 2,500 targeted songs. For class k with count c_k, quota Q_k

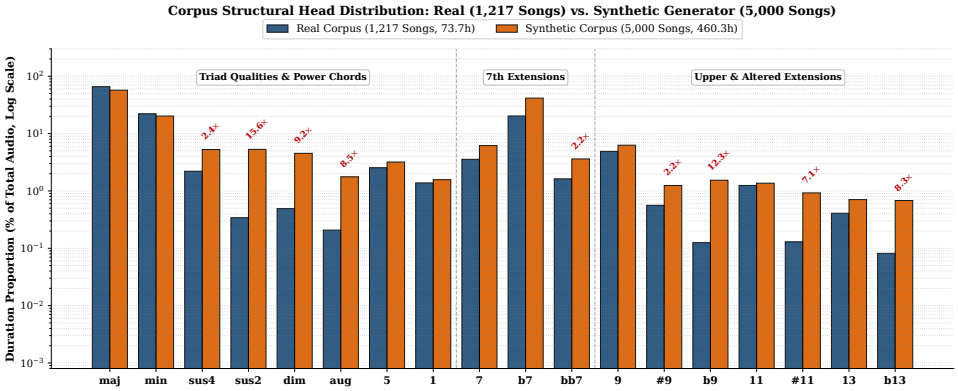


Figure 1. Log-scale duration shares across 18 chord and extension classes in the 1,217-song real benchmark (73.7 h) [5] versus our 5,000-song synthetic corpus (460.3 h, excluding `N`). Major (65.75%) and minor (22.16%) triads occupy 87.91% of real audio, leaving a steep Zipfian tail; quota scheduling expands rare triads (`sus2`: 0.34%→5.31%, 15.6×; `dim`: 9.2×; `aug`: 8.5×; `sus4`: 2.4×) and upper extensions (`b9`: 12.3×; `b13`: 8.3×; `#11`: 7.1×; `#9`: 2.2×).

($S_{\text{triad}} = 3.0\times$ on rare triads), event budget N_{ev}, and buffer $B \in \{50, 5{,}000\}$, due positions satisfy:

$$\text{pos}_{\text{due}}(k) = \left\lfloor \frac{(c_k + 1)N_{\text{ev}} + (Q_k + B) - 1}{Q_k + B} \right\rfloor, \quad (3)$$

spaced by $\Delta_{\text{pos}} \geq 2$ along attested 1- or 2-step edges weighted by $P_{\text{dist1}}(s)C_{\text{local}}(s)/(1 + \text{RootCount}(s))$ (3.38×–95.8× empirical upweighting). Versus the 1,217-song corpus (Figure 1), duration shares rise across `sus2` (0.34%→5.31%, 15.55×), `dim` (0.49%→4.53%, 9.20×), `aug` (0.21%→1.77%, 8.50×), `sus4` (2.21%→5.28%, 2.40×), `b7` (20.30%→41.55%), `bb7` (1.62%→3.61%), `b9` (0.13%→1.54%, 12.28×), `b13` (0.08%→0.68%, 8.34×), `#11` (0.13%→0.92%, 7.12×), `#9` (0.56%→1.25%, 2.22×), `9` (4.90%→6.27%), `13` (0.41%→0.71%), and `11` (1.25%→1.36%).

3.2 Symbolic Realization and Voicing Heuristics

Using `render.py`, we arrange progressions onto a 4/4 16th-note grid (4 slots/beat) across four JFugue tracks: `V0` (70% pads, 30% arpeggios), `V1` (bass in $[28, 55]$, E_1–G_3, playing roots, inversions, walking/passing lines, and $p_{\text{jump}} = 0.20$ octave leaps), `V2` (beam-search or random-walk melody masking $\{3, 4\}$ on `sus2`/`sus4`/`1` and restricting `5` to $\{0, 5, 7, 10\}$), and `V9` (Channel 10 drums).

Six policies parameterize `V0` MIDI bounds $[w_{\text{lo}}, w_{\text{hi}}]$ and voice limits $[V_{\min}, V_{\max}]$ [20]: `pop-piano` ($[43, 88]$, $[3, 7]$), `pop-guitar` ($[40, 76]$, $[3, 6]$), `pop-synth` ($[36, 96]$, $[3, 8]$), `jazz-piano` ($[48, 84]$, $[3, 6]$), `jazz-guitar` ($[45, 79]$, $[3, 5]$), and `jazz-synth` ($[40, 91]$, $[3, 7]$), with guitar shapes enforcing $\leq$ 4-fret spans (cap $\leq$ 12/15) for seventh-chord acoustics [21]. Unlike shell voicers that drop extensions, `voicing/select.py` enforces acoustic faithfulness: (1) 100% of 7th–13th extensions sound in piano/synth policies, while guitar policies allow fallback only below Step 8 of an 8-level ladder at $+60.0N_{\text{dropped}}$ penalty ($< 0.5\%$ drop rate); (2) the Distinguishing Chord Tone (13th > 11th > 9th > 7th > 3rd > 5th) avoids upper-root masking via top, isolated, or octave exposure; (3) 3rds never drop and 5ths omit ($p_{\text{omit_5th}} \in [0.10, 0.70]$) only on `maj`/`min`; and (4) `V0` root omission ($p_{\text{omit_root}} \in [0.02, 0.75]$) requires active `V1` bass. After low-interval and cluster pruning, we sample voicings via $P(c) \propto \exp(-(C(c) - \min C)/\tau_{\text{voic}})$ ($\tau_{\text{voic}} \in [2.2, 6.0]$) under:

$$\begin{aligned} C(c) = {} & w_{\text{vl}} d_{\text{vl}}(p_{\text{prev}}, p_c) - B_{\text{trans}} \\ & - 0.5 \min(|p_{\text{prev}} \cap p_c|, 3) + C_{\text{reg}}(c), \end{aligned} \quad (4)$$

combining Hungarian voice-leading distance d_{vl} [22, 23] ($w_{\text{vl}} \in [0.5, 1.6]$, unmatched penalty $[1.2, 2.5]$), Neo-Riemannian P/L/R bonus $B_{\text{plr}} \in [0.4, 3.0]$ [24, 25], and spacing/drift cost C_{reg}. Five arpeggio profiles (gate 0.65–1.75, velocity 78–96) and `HumanizedMidiRenderer.java` prevent block-onset overfitting via $\sigma_t = 15.0$ ms jitter, pocket offsets $(0, +8, -4, -6, +2$ ms on pads, `V1`, `V2`, arpeggios, `V9`), 7.0 ms/string strums (GM 24–31), CC64 sustain (GM 0–7), and Gaussian velocity noise ($\sigma_v = 8.0$ harmonic; $\sigma_v = 4.0$ atop drum hits $\mathcal{N}(100, 8^2)$ kick/snare/crash, $\mathcal{N}(65, 8^2)$ ghost snare, $\mathcal{N}(80, 12^2)$ hi-hat/ride).

3.3 Synthesis and Random-Mix Augmentation

We synthesize stems via FluidSynth at $f_s = 22{,}050$ Hz (16-bit FLAC) [11], rotating across four GM `.sf2` banks: `FluidR3_GM` (141 MB), `MuseScore_General_Full` (468 MB), `GeneralUser_GS` (31 MB), and `TimGM6mb` (5.7 MB). GM pools span 40 `V0` patches (16 keyboard/organ/mallet, 12 guitar/plucked, 12 synth/pad), 13 weighted `V1` bass patches (GM 33, 34, 32, 35 weighted 35, 25, 20, 15), 18 `V2` lead patches ($\geq$ 25% vocal formants GM 52–54 in pop/rock), and randomized drum kits ($p = 0.60$ crash).

To mitigate acoustic overfitting, we mix stems via `project_condition_corpus.py` using Bernoulli dropout on `V2` and `V9` ($P = 0.70$ each: 49% four-stem, 21% `cbm`, 21% `cbp`, 9% `cb`) and perturb nominal gains $(0.0, +1.0, -1.0, -1.5)$ dB by $\Delta G_{\text{V0}}, \Delta G_{\text{V1}} \sim U(-3, +3)$ dB, $\Delta G_{\text{V2}} \sim U(-3, +4)$ dB, and $\Delta G_{\text{V9}} \sim U(-4, +4)$ dB. FFmpeg applies `V1` saturation ($x - 0.05x^3$), 120 Hz shelf (+2.0 dB), and 3.5 kHz low-pass; `V0` power-chord overdrive and cabinet EQ ($-3.0, +2.5, -6.0$ dB at $0.1, 0.25, 4.5$ kHz); stereo

panning (60/40 `V0`, 50/50 `V2`, 40/60 `V9`); split-band room convolution (1.2/0.8/0.4 s, 82%/18% dry/wet, 150 Hz wet high-pass); detuning $\Delta f_{\text{cents}} \sim U(-25, +25)$ cents; $\pm U(1.5, 3.0)$ dB tilt at 1.5 kHz; console saturation ($x - 0.1x^3$); bus compression (20/150 ms); and -18.0 LUFS normalization (-1.0 dBTP ceiling). This forces the Conformer [4, 11] to extract timbre-invariant pitch-class profiles across polyphonic masking and detuned CQT bins. Across all 5,000 tracks (460.3 h), `start_padding_frames=0` anchors sample 0 to $t = 0.000$ s, exporting frame-exact Harte `.lab` labels [15] free of human lag [8].

4. EXPERIMENTAL SETUP

4.1 Audio Preprocessing and ChordFormer Architecture

Audio is downmixed to mono at $f_s = 22{,}050$ Hz [11]. We extract hybrid CQT spectrograms [26] with hop $H = 512$ samples ($\Delta t = 23.22$ ms), window $N_{\text{win}} = 8{,}192$ samples, and $b = 36$ bins/octave (33.3 cents/bin) spanning $F_{\text{raw}} = 288$ bins (8.0 octaves) from $f_{\min} = 23.12$ Hz ($\text{F}\sharp_0$, $A_4 = 440.0$ Hz). Online pitch shifting steps across $[-5, +6]$ semitones on a 3-bin grid, slicing $F = 252$ bins (7.0 octaves, $\text{C}_1 = 32.70$ Hz to $\text{B}_7 = 3{,}951.07$ Hz). Log-amplitude scaling clips to $[-80.0, 0.0]$ dB, and right-zero-padding by 4 bins yields $d_{\text{in}} = 256$ channels ($\mathbf{x} \in \mathbb{R}^{B\times T\times 256}$).

In `confor_head16.py`, ChordFormer [11] stacks $N_L = 4$ Macaron Conformer blocks [4] ($d_{\text{model}} = 256$, $p_{\text{drop}} = 0.10$): (1) half-step FFN_1 ($d_{\text{ffn}} = 1{,}024$); (2) depthwise convolution ($K = 31$ frames ≈ 720 ms, `SAME` pad 15, Pointwise $256 \to 512 \to$ GLU, Depthwise 256, 1-group GroupNorm, SiLU, Pointwise $256 \to 256$, `convolution_first=True`); (3) 16-head MHSA ($h = 16$, $d_k = 16$); (4) half-step FFN_2; and (5) LayerNorm. A linear head maps $\mathbf{h}_t \in \mathbb{R}^{256}$ to $\hat{\mathbf{z}}_t = [\hat{\mathbf{z}}^{(97)}_{0,t} \mid \hat{\mathbf{z}}^{(13)}_{1,t} \mid \hat{\mathbf{z}}^{(4)}_{2,t} \mid \hat{\mathbf{z}}^{(4)}_{3,t} \mid \hat{\mathbf{z}}^{(3)}_{4,t} \mid \hat{\mathbf{z}}^{(3)}_{5,t}] \in \mathbb{R}^{124}$ (Section 3.1) [10, 11], totaling $|\Theta| = 6{,}123{,}644$ parameters.

During inference (`extractors/xhmm_ismir.py`), posteriors $\mathbf{p}_{h,t} = \text{Softmax}(\hat{\mathbf{z}}_{h,t})$ are decoded via linear-chain CRF Viterbi smoothing over Harte templates [11, 15]:

$$S^*(t) = \arg\max_c \Big\{ \max_{c'} \left(V(t-1, c') - \tau_{\text{crf}} \cdot \mathbb{I}(c' \neq c)\right) + \log P(c|\mathbf{x}_t) \Big\}, \tag{5}$$

with $\tau_{\text{crf}} = 2.0$ (multi-head) and $\tau_{\text{crf}} = 30.0$ (MIREX). Fusing the 5-fold ensemble into FP16 TensorRT 10.x on Google Cloud GKE (NVIDIA L4, 24 GB) with dual-stream dynamic batching in Triton achieves sub-40 ms latency per 10.0 s chunk (431 frames): 2.83 ms compute at $L = 1$, and 10.93 ms compute (32.56 ms round-trip) at $L = 13$ peak throughput (395.75 inf/s, 4.28× speedup).

4.2 Datasets and Cross-Validation Protocol

We evaluate across two corpora: (1) **Synthetic Pretraining Corpus**, comprising 5,000 procedural random-mix tracks from Section 3 (460.3 h, 21.8 GB CQT HDF5), split into 4,500 training and 500 validation tracks (seed `42`); and (2) **Real Benchmark Corpus**, the 1,217-song collection of Humphrey and Bello [5] spanning 731 McGill Billboard [6], 287 USPop2002/MARL, and 199 Isophonics (Beatles/Queen) tracks [7]. Under 5-fold cross-validation (seed `20190326`), mutually exclusive splits `s0`–`s4` (730/731 train, 243 val, 243/244 full-song test tracks/fold) total 11,436,357 evaluated frames (73.7 h). In Fold 0, majority `maj` (4,465,981 frames) and `min` (1,586,913) outnumber `sus4` (146,527, 30.5 : 1), `dim` (33,534, 133.2 : 1), `sus2` (20,531, 217.5 : 1), and `aug` (15,272, 292.4 : 1), while upper extensions drop from `9` (333,926) to `#9` (27,864, 221.8 : 1), `b9` (8,763, 705.2 : 1), `#11` (5,804, 1,113.1 : 1), and `b13` (5,522, 1,159.3 : 1).

4.3 Training Regimens: Baseline vs. Sim-to-Real Curriculum

We compare two regimens under FP16 mixed precision and AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$) on $T = 1{,}000$-frame chunks (23.22 s): (1) **Control Baseline** (`pretrained-benchmark`), trained from scratch on the 1,217 real songs for 24 epochs/fold ($B = 36$ or 48, uniform LR η over $\{10^{-3}\!:\!12, 10^{-4}\!:\!8, 10^{-5}\!:\!4\}$ epochs, $[-5, +6]$ pitch shifting); and (2) **Sim-to-Real Curriculum** (`random-mix-finetuned-second-benchmark`), pretraining in Stage 1 (`pretrain_synthetic.py`) on 5,000 synthetic tracks for 7 epochs (10,500 mini-batches, $B = 36$, LR $\{10^{-3} : 4, 10^{-4}\!:\!2, 10^{-5}\!:\!1\}$) with unweighted cross-entropy, pitch shifting, and SpecAugment [27] (2 frequency masks $\leq$ 24 bins, 2 time masks ≤ 60 frames at -80.0 dB), then finetuning in Stage 2 (`finetune_real_fold.py`) on each real fold for 24 epochs ($B = 36$, LR $\{10^{-4} : 12, 10^{-5} : 8, 10^{-6}\!:\!4\}$).

Real training minimizes $\mathcal{L}_{\text{total}} = \sum_{h=0}^{5} \lambda_h \mathcal{L}_h(\hat{\mathbf{z}}_h, \mathbf{y}_h; \mathbf{w}_h)$ ($\boldsymbol{\lambda} = [1, 1, 1, 1, 1, 1]$) with inverse-frequency class weights [10, 11]:

$$w_{h,c} = \min\left(\left(\frac{\max_k N_{h,k}}{\max(N_{h,c}, 1.0)}\right)^{\alpha}, C_{\max}\right), \tag{6}$$

using $\alpha = 0.50$, $C_{\max} = 10.0$, and clamps ($w_{2,\text{b7}} \approx 3.602$, $w_{3,9} = 6.000$, $w_{\text{sus4}} \geq 6.800$, $w_5 \geq 6.000$, $w_{\text{min}} \geq 2.000$). Stage 2 adds: (1) **Inverted Discriminative LR** ($\mathbf{s}_{\text{lr}} = [0.2, 0.5, 1.0]$), scaling bottom blocks (Layers 0–1) by 0.2η, top blocks (Layers 2–3) by 0.5η, and output heads by 1.0η; and (2) **Floor Curriculum Replay**, injecting SpecAugmented synthetic chunks at $\rho(\eta) = 0.150$ ($\eta \geq 10^{-4}$), 0.125 ($\eta = 10^{-5}$), and a strict floor $\rho_{\min} = 0.100$ ($\eta \leq 10^{-6}$) against catastrophic forgetting.

Table 1. 5-fold macro recall and F_1 (%) across the 6 structural classification heads and isolated triad quality subsets (mean ± std across splits s_0–s_4). Control: 1,217-song real baseline; Experimental: 5,000-song synthetic pretraining + real finetuning (Δ = Exp. − Control).

Structural Output Head	Control Rec. (%)	Exp. Rec. (%)	Δ Rec.	Control F_1 (%)	Exp. F_1 (%)	Δ F_1
Primary 6-Head Multi-Task Decomposition:						
Root + Triad (97 classes)	41.21 ± 3.12	**42.66 ± 2.59**	+1.45	36.57 ± 2.07	**36.70 ± 1.68**	+0.13
Bass Pitch (13 classes)	**84.81 ± 1.73**	82.93 ± 1.51	-1.88	**84.72 ± 1.77**	83.01 ± 1.57	-1.70
7th Extension (4 classes)	**62.69 ± 4.27**	62.55 ± 3.14	-0.14	**57.67 ± 2.73**	56.99 ± 1.48	-0.67
9th Extension (4 classes)	35.33 ± 2.68	**43.58 ± 7.54**	+8.25	34.98 ± 2.96	**39.01 ± 6.25**	+4.03
11th Extension (3 classes)	38.74 ± 7.85	**41.71 ± 8.03**	+2.97	37.35 ± 3.60	**42.14 ± 7.04**	+4.79
13th Extension (3 classes)	**35.36 ± 3.74**	34.26 ± 1.95	-1.10	**36.09 ± 4.82**	34.47 ± 2.47	-1.62
Macro Average (6 Heads)	49.69 ± 2.10	**51.28 ± 1.94**	+1.59	47.90 ± 1.69	**48.72 ± 1.86**	+0.82
Targeted Triad Quality Breakdown:						
Triad Quality (all 9 classes)	50.23 ± 4.53	**53.42 ± 3.96**	+3.19	45.52 ± 3.40	**46.91 ± 2.57**	+1.39
Common Triads (*maj*, *min*)	**84.74 ± 1.97**	83.08 ± 1.44	-1.66	**86.99 ± 1.19**	86.05 ± 0.85	-0.94
Rare Triads (*sus2*, *sus4*, *dim*, *aug*)	41.67 ± 7.30	**48.31 ± 6.47**	+6.64	53.11 ± 8.62	**60.94 ± 6.26**	+7.83

5. RESULTS AND DISCUSSION

5.1 Structured Head-Wise Recall and Rare-Class Gains

Table 1 summarizes 5-fold cross-validated performance ($\tau_{\text{crf}} = 2.0$) across the six structural heads over all 11,436,357 test frames. Across the 6-head macro average, our Sim-to-Real curriculum raises recall from $49.69 \pm 2.10\%$ to $51.28 \pm 1.94\%$ (+1.59 pp) and macro F_1 from $47.90 \pm 1.69\%$ to $48.72 \pm 1.86\%$ (+0.82 pp), concentrating gains on sample-starved classes [1, 5].

On Head 0 (Root + Triad, 97 classes), macro recall increases from $41.21 \pm 3.12\%$ to $42.66 \pm 2.59\%$ (+1.45 pp) and F_1 from $36.57 \pm 2.07\%$ to $36.70 \pm 1.68\%$ (+0.13 pp), driven by non-diatonic structures: overall triad recall rises from $50.23 \pm 4.53\%$ to $53.42 \pm 3.96\%$ (+3.19 pp; F_1 $45.52 \pm 3.40\% \rightarrow 46.91 \pm 2.57\%$). While common triads (`maj`, `min`) shift by −1.66 pp in recall ($84.74 \pm 1.97\% \rightarrow 83.08 \pm 1.44\%$) and −0.94 pp in F_1 ($86.99 \pm 1.19\% \rightarrow 86.05 \pm 0.85\%$), rare triads (`sus2`, `sus4`, `dim`, `aug`) surge from $41.67 \pm 7.30\%$ to $48.31 \pm 6.47\%$ in recall (+6.64 pp, +15.9% relative) and from $53.11 \pm 8.62\%$ to $60.94 \pm 6.26\%$ in F_1 (+7.83 pp, +14.7% relative). Similarly, Head 3 (9th) recall jumps from $35.33 \pm 2.68\%$ to $43.58 \pm 7.54\%$ (+8.25 pp, +23.4% relative; F_1 +4.03 pp to $39.01 \pm 6.25\%$), and Head 4 (11th) rises from $38.74 \pm 7.85\%$ to $41.71 \pm 8.03\%$ (+2.97 pp; F_1 +4.79 pp to $42.14 \pm 7.04\%$).

Per-class metrics (Figure 2) confirm large gains across rare qualities and extensions: `aug` recall rises from 43.37% to 61.02% (+17.65 pp, F_1 35.40% → 43.15%), `sus2` from 17.32% to 30.25% (+12.93 pp, F_1 17.40% → 19.90%), `5` from 28.76% to 41.91% (+13.15 pp, F_1 28.49% → 29.50%), `dim` from 65.61% to 67.97% (+2.36 pp, F_1 41.65% → 47.79%), `sus4` from 51.19% to 53.72% (+2.53 pp), and `b7` from 70.31% to 80.90% (+10.59 pp; `bb7` F_1 23.15% → 25.22%). On upper extensions, natural `9` leaps by +25.16 pp (31.64% → 56.80%, F_1 31.06% → 36.19%); `#9` rises 79× (0.07% → 5.53%, F_1 0.10% → 4.97%); `b9` rises +6.32 pp (18.39% → 24.71%, F_1 9.56% → 18.02%); `#11` rises +8.87 pp (14.15% → 23.02%, F_1 13.39% → 20.50%); and `b13` jumps 7.8× (0.84% → 6.58%, F_1 0.77% → 8.43%).

These gains cannot be explained by loss reweighting alone. Both models use identical inverse-frequency reweighting ($\alpha = 0.50, C_{\max} = 10.0$) in real training [10, 11]; yet without synthetic pretraining, amplifying gradients on a handful of real tracks magnifies gradient variance and overfits to track-specific timbres. Physically voiced synthetic pretraining provides the spectral support needed for best-in-class recall across large-vocabulary architectures [9–11].

Table 2. Weighted Chord Symbol Recall (WCSR, %) across 5-fold cross-validation on 1,217 real songs (s_0–s_4, mean ± std). Control: 1,217-song real baseline; Experimental: synthetic pretraining + real finetuning.

Evaluation Rule	Control (%)	Experimental (%)	Δ (pp)
Root Pitch Only	**83.77 ± 1.88**	82.30 ± 1.58	-1.47
Major / Minor	**79.78 ± 2.38**	77.04 ± 1.71	-2.74
Major / Minor + Inversions	**77.77 ± 2.22**	75.17 ± 1.71	-2.60
Triads (Full Quality)	**75.01 ± 2.72**	72.88 ± 1.97	-2.13
Triads + Inversions	**73.15 ± 2.58**	71.13 ± 1.95	-2.02
Sevenths (Standard)	**66.46 ± 2.46**	62.46 ± 1.73	-4.00
Sevenths + Inversions	**65.12 ± 2.42**	61.18 ± 1.72	-3.94
Tetrads (Extended)	**61.62 ± 2.73**	58.20 ± 1.74	-3.42
Tetrads + Inversions	**60.38 ± 2.69**	57.02 ± 1.75	-3.36
MIREX Metric (Overall)	**81.49 ± 2.56**	79.59 ± 1.85	-1.90

5.2 Standard Frame-Level WCSR and MIREX Evaluation

Table 2 reports duration-weighted chord symbol recall (WCSR) via `mir_eval.chord` [28] ($\tau_{\text{crf}} = 30.0$). Across all ten comparison rules, the Sim-to-Real curriculum scores strictly lower than the real-only baseline: Root drops by 1.47 pp (83.77% → 82.30%), Major/Minor by 2.74 pp (79.78% → 77.04%; −2.60 pp with inversions), Triads by 2.13 pp (75.01% → 72.88%; −2.02 pp with inversions), Sevenths by 4.00 pp (66.46% → 62.46%), Tetrads by 3.42 pp (61.62% → 58.20%), and overall MIREX WCSR by 1.90 pp ($81.49 \pm 2.56\% \rightarrow 79.59 \pm 1.85\%$).

This divergence reflects a fundamental precision-recall and metric-weighting trade-off [1, 5]. WCSR weights accuracy strictly by interval duration ($\sum_i d_i \mathbb{I}(\hat{C}_i == C_i) / \sum_i d_i$) [28], biasing it toward `maj` (44.93% of real duration) and `min` (11.95%), with unextended triads exceeding 70% of frames. Under Harte recombination, candidate log-probabilities sum across all six heads ($\log P_k(t) = \log p_{\text{triad}} + \mathbb{I}(\text{use_bass}) \log p_{\text{bass}} + \sum_e \log p_e$). Expanding minority decision boundaries in $\hat{\mathbf{z}}_{h,t}$ to capture rare qualities and extensions inevitably introduces false-positive activations on acoustically ambiguous frames of

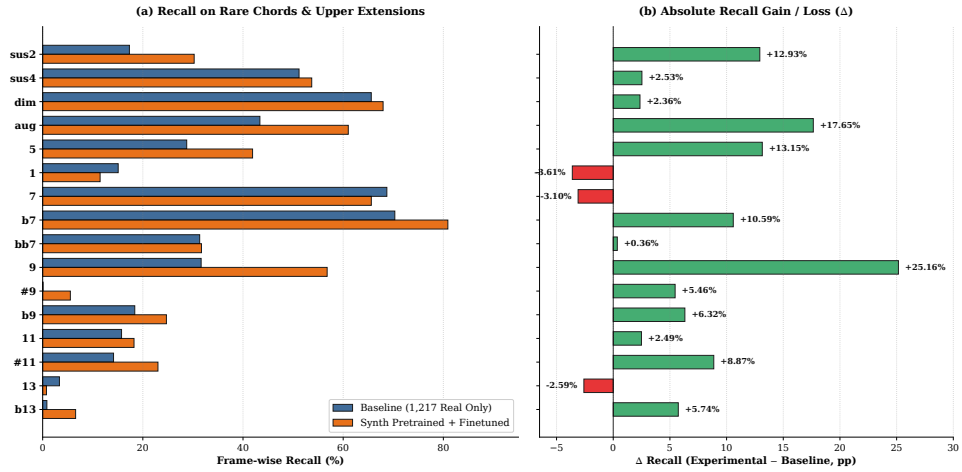


Figure 2. Per-class evaluation on rare triads, power chords, 7th extensions, and upper/altered extensions across 11,436,357 frames of the 1,217-song real benchmark: (a) absolute recall comparing Control Baseline (real-only) versus Experimental Curriculum (synthetic pretraining + real finetuning); (b) diverging recall gain (ΔRecall), highlighting gains across `9` (+25.16 pp), `aug` (+17.65 pp), `5` (+13.15 pp), `sus2` (+12.93 pp), `b7` (+10.59 pp), `#11` (+8.87 pp), `b9` (+6.32 pp), `b13` (+5.74 pp), and `#9` (+5.46 pp).

dominant triads (e.g., guitar overtones or vocal passing tones triggering `sus4` or `b7`, where per-class F_1 shifts by -1.79 pp and -0.89 pp despite $+2.53$ pp and $+10.59$ pp recall gains). Because rare classes occupy minuscule duration slices (`sus2` 0.34%, `aug` 0.21%, `b9` 0.13%, `b13` 0.08%), true-positive detections on the harmonic tail yield negligible WCSR credit, whereas any false-positive excursion on a majority triad immediately penalizes duration overlap.

Stage 2 adaptation bounds this trade-off. Evaluating synthetic models without real finetuning degrades MIREX WCSR by $>$ 15 pp [19]. Updating linear heads at 1.0η while anchoring bottom Conformer blocks (0.2η) and maintaining the $\rho_{\min} = 0.10$ synthetic replay floor recalibrates output logit priors to studio mixes without erasing minority features (while fourth-root loss calibration $\alpha = 0.25, C_{\max} = 4.0$ can further prioritize majority precision). Moreover, synthetic pretraining reduces 5-fold standard deviation across all ten rules in Table 2 (MIREX $\pm 2.56\% \rightarrow \pm 1.85\%$; MajMin $\pm 2.38\% \rightarrow \pm 1.71\%$; Tetrads $\pm 2.73\% \rightarrow \pm 1.74\%$), confirming improved cross-fold stability.

5.3 Sim-to-Real Acoustic Domain Gap and Extension Masking

Bridging SoundFont synthesis to studio recordings requires overcoming static synthesis timbres and polyphonic overtone masking. Raw SoundFont waveforms exhibit deterministic attack transients and unvarying harmonic envelopes that Conformer encoders easily memorize [19]. Our random-mix pipeline counters this via Bernoulli stem dropout ($P = 0.70$ on lead/drums), gain perturbations ($\Delta G \in [-4, +4]$ dB), cabinet overdrive, split-band room convolution, and continuous detuning ($\Delta f_{\text{cents}} \sim U(-25, +25)$ cents), forcing the $K = 31$ depthwise convolutions and self-attention heads to extract timbre-invariant pitch-class intervals across detuned CQT bins.

Furthermore, Figure 2 shows how physical register and overtone alignment govern extension recognition: (1) *Low-Register Triads*: Rare triads (`sus2`, `aug`, `dim`) alter third/fifth degrees in C_1–C_4, where 36 bins/octave CQT filterbanks resolve fundamentals cleanly, yielding uniform recall (+2.36 pp to +17.65 pp) and F_1 gains; (2) *Natural Upper Extensions* (`11`, `13`): In C_4–B_6, these tones compete with vocal formants, guitars, and cymbals. Natural `11` (17 semitones) clashes against a major 3rd and is confounded with passing tones (+2.49 pp recall), while natural `13` (21 semitones) drops in recall ($3.35\% \rightarrow 0.76\%$) because its partials align with root and fifth overtones; and (3) *Altered Extensions* (`#9`, `b9`, `#11`, `b13`): Chromatic alterations diverge from the diatonic overtone series of the triad, producing isolated CQT peaks that physically voiced pretraining detects (79× on `#9`, 7.8× on `b13`).

6. CONCLUSION

We presented a Sim-to-Real pretraining framework for large-vocabulary audio chord recognition that synthesizes 5,000 multitracks (460.3 h) via empirical transition sampling, rare-chord quota scheduling, physically constrained voicings, and random-mix augmentation. Pretraining ChordFormer on synthetic audio and finetuning with inverted discriminative learning rates ($\mathbf{s}_{\text{lr}} = [0.2, 0.5, 1.0]$) and floor curriculum replay ($\rho_{\min} = 0.10$) achieves best-in-class recall across rare triads (+6.64 pp), natural 9ths (+25.16 pp), and alterations (79× on `#9`, 7.8× on `b13`) on the 1,217-song benchmark while reducing 5-fold WCSR variance. Code and models are at `https://github.com/anonymous/probabilistic-composition-generator` and `https://github.com/anonymous/ChordFormer-Artificial-Dataset-Benchmarking`. Because our 5,000-track corpus tests one operating point in the generator's quota space, future work will optimize class proportions for lagging degrees (natural 13ths, `1`) and integrate differentiable synthesis (DDSP).

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